Ahmed Younis at ImageEval 2026 Shared Tasks: The Judge Needs a Prior and the Task Needs a Constraint

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Abstract

Across two ImageEval 2026 tracks, our best results came from analyzing the evaluation machinery rather than from a stronger model, yielding 1st place on hallucination detection (Task 1b, Modern Standard Arabic) and 4th on CRAI-Bench (Task 2). On CRAI-Bench, which asks systems to predict human cultural-accuracy scores for AI-generated images of Qatari scenes, a *blind frequency prior* that never looks at an image reaches dev Spearman 0.6878, above the organizers' GPT-4-as-judge baseline of 0.625; adding two vision judges, with blend weights tuned against the global ranking metric rather than by scene-level cross-validation, reaches **0.7806 on the blind test**. The weighting correction alone is worth +0.0296 test Spearman, because scene-level cross-validation is structurally blind to the within-cell ranking that judges contribute. On Task 1b, reframing "label three statements true/false" as "select the one grounded statement" makes invalid labelings structurally impossible and reaches **0.964 combined accuracy on the blind test**, against an official baseline of 0.508. A matched-model ablation shows most of that headline gap is the underlying judge, but the constraint still adds +0.026 combined accuracy and eliminates the 4.0% of items that otherwise receive structurally invalid labelings; meanwhile upgrading to a stronger judge is worth nothing measurable and a three-model vote scored lower.

1 Introduction

ImageEval 2026 (Abdaljalil et al., 2026) poses two Arabic multimodal evaluation problems.[1] **CRAI-Bench** gives a reference photograph of an authentic Qatari scene, a caption, and an image generated from that caption, and asks for five Cultural Representation Accuracy Index scores plus a composite; systems are ranked by Spearman correlation with human annotators. **Task 1b** gives an image and three Modern Standard Arabic (MSA) statements of which exactly one is grounded, and asks for a true/false label on each.

Both tasks are meta-evaluation: the system under test is itself a judge. This makes two questions unusually load-bearing (*what floor should a judge be measured against*, and *what does the metric actually reward*), and our systems are organized around them:

- A frequency prior over the benchmark's own caption-ablation structure sets a floor that the official LLM-judge baseline does not clear. We frame this prior as a *benchmark diagnostic*, not a cultural-understanding system (§3.2).
- Judge weights chosen by scene-level cross-validation are structurally blind to within-cell ranking; re-tuning them against the global ranking metric is worth +0.0296 test Spearman (§3.3).
- On Task 1b, the "exactly one is true" constraint should be enforced by the decoding formulation, not hoped for from the model; a matched-model ablation prices the constraint at +0.026 combined accuracy plus a structural validity guarantee (§4).

2 Related Work

Using LLMs as judges of other systems' outputs is now standard practice (Zheng et al., 2023; Liu et al., 2023), and its reliability is typically audited by correlating judge scores with human gold. Our contribution to that audit methodology is a cheaper sanity floor: a metadata-only prior that any judge must beat before its scores can be attributed to

[1] We competed as team *Ahmed Younis*, Codabench username `ahmedbakr`. Official final runs: Task 2 the globally-weighted blend of §3.3; Task 1b submission 884266 (`gemini-3.1-pro-preview` selection decode). Code and prompts: `github.com/itsbakr/imageeval-2026`.

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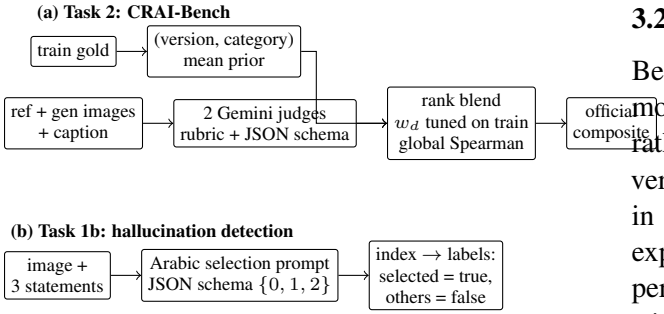


Figure 1: System pipelines. (a) A blind frequency prior and two vision judges are combined per dimension in rank space; weights are tuned on train against the global ranking metric, and the official composite formula produces the submission. (b) The 1b decoder selects the single grounded statement and expands the index into the task's required per-statement true/false labels, making invalid labelings impossible. Verbatim prompts are in Appendix A.

reading the inputs at all (§3.2, §3.6). The tracks build on the organizers' line of work on culturally grounded Arabic multimodal evaluation: everyday visual question answering (Alam et al., 2025), counterfactual hallucination suites (Mousi et al., 2026b), and cross-modal instability measurement (Mousi et al., 2026a), which CRAI-Bench extends from recognition to generation evaluation. Our companion ArGuard system paper makes the same argument for the decision layer of classification tasks: calibration and decoding, not model choice, dominated the outcome (Younis, 2026).

3 Task 2: CRAI-Bench

3.1 Data and the structure that matters

The release provides 120 train, 40 dev and 40 test instances. Each instance is one of five caption *versions* of a scene, ordered from v1 (fully culturally specified) to v5 (culturally stripped), paired with a generated image and a reference photograph, with human gold on five dimensions: Cultural Element Accuracy (CEA), Contextual Coherence (CC), Cultural Specificity (CS), Cultural Integrity (CI) and Hallucination Penalty (HP). The composite is a fixed weighted sum; for reproducibility, we take the exact weights from the official format checker, which is authoritative for scoring (CEA 0.30, CC 0.20, CS 0.20, CI 0.20, HP -0.10, with CS restricted to $\{0, 0.25, 0.5, 0.75, 1\}$ and composite consistency checked to 0.01).

3.2 A blind prior sets the floor

Because the five versions form a designed ablation, most of the rankable variance is between versions rather than between images: on train gold, per-version means alone explain 67.7% of the variance in the composite, and (version, category) means explain 69.3%. Our first system exploits only this: per-(version, category) means estimated on train, with the composite recomputed through the official formula and CS snapped to a legal tier. It reads no images, calls no model, and reaches **dev Spearman 0.6878** (MAE 0.1927) against the official GPT-4-judge baseline of 0.625.

This is a statement about the benchmark, not about our system: the prior is a *diagnostic floor*, not a component that demonstrates visual understanding. Benchmarks constructed as systematic caption ablations are substantially solvable from metadata, so a judge that merely beats random has not been shown to be reading the images at all. We recommend reporting this floor alongside judge results.

3.3 Judges, weighted against the ranking metric

Our judges see the reference image and the generated image together, so "culturally accurate" is posed as a comparison rather than an absolute. One call returns all five dimensions with a per-dimension rubric (Appendix A); CS is emitted directly as one of the five legal tiers; and HP is a *checklist*: the judge must name each fabricated element, and HP is $\min(0.1 \times \text{count}, 1)$ rather than a free-floating score. We use `gemini-3.1-pro-preview` and `gemini-3.6-flash` at temperature 0 with a server-side JSON schema, which we found necessary: without it the pro model returned unparseable output on roughly 30% of calls (27/40 items succeeded; 40/40 with the schema).

Because Spearman rewards ranking, we blend in rank space per dimension, $(1 - w_d) \cdot \text{rank}(\text{prior}) + w_d \cdot \text{rank}(\text{judge mean})$, with w_d chosen per dimension on train only; the blended ranks are mapped back into each dimension's train value range before the composite is recomputed (the full procedure, including the coordinate-descent grid search, is specified in Appendix B). The dev set is used only as a final out-of-sample check, never for weight selection.

Our first weighting chose w_d by leave-one-

scene-out cross-validation, which selects $w = 0$ for CEA, CC, CI and HP and $w = 0.3$ for CS alone; apparently, judges add almost nothing. This turned out to be an artifact of the CV design, and correcting it is the paper's central methodological finding. Scene-level CV scores ranking *within* a scene's five versions, where the prior is already strong; but the task metric is *global* Spearman, and every dev item lies in a (version, category) cell where the prior emits tied scores. Ranking within those ties is precisely what a judge can do, and scene-level CV cannot see it. Re-tuning the weights against global composite Spearman on train (dev untouched) selects moderate judge weights (CEA 0.3, CC 0.4, CS 0.6, CI 0; HP's tuned weight is moot because its gold is degenerate, §3.5) and improves out-of-sample dev Spearman from 0.7076 to 0.7454, and blind-test Spearman from 0.7510 to **0.7806**.

3.4 Results and component analysis

Table 1 reports each component alone and blended. Three readings matter. First, the blind prior clears the official judge baseline without reading an image. Second, each judge alone also lands near the prior (`flash` 0.7206, `pro` 0.6736 dev; the cheaper judge is the stronger one here, and the two tie in §4), so neither signal dominates: the gain comes from blending them against the right objective. Third, on robustness: a paired bootstrap over the 40 dev items puts the re-weighting gain at $+0.05$ with 95% CI $[-0.02, +0.15]$, i.e. not individually significant at $n = 40$, which is why we treat the +0.0296 improvement on the disjoint blind test as the out-of-sample confirmation rather than the dev delta alone.

Where the blend still fails. The five largest dev errors are all architecture-category scenes whose gold sits at the extremes (composites of 0.05 or 0.90): the prior regresses these toward their cell mean, and the judges, whose weight is moderate, do not override it. The residual failure mode is therefore compression of extremes, not misordering of typical items, which is consistent with a rank-space blend anchored on cell means.

3.5 HP is degenerate as annotated

Human HP gold is near-constant: our test HP MAE is 0.0083 at $\rho = 0.0000$, i.e. we predict the right values but there is no variance to correlate against. The organizers' own GPT-4 judgments correlate *negatively* with human HP (-0.120). As currently annotated, HP cannot discriminate between systems, and its -0.10 weight in the composite mostly injects noise. We suggest either annotating hallucination at higher granularity or removing it from the ranking metric.

System	dev ρ	test ρ
Official GPT-4 judge baseline	0.6250	–
Blind version prior (no images)	0.6878	–
Judge only: `gemini-3.6-flash`	0.7206	–
Judge only: `gemini-3.1-pro`	0.6736	–
Judge only: mean of both	0.7026	–
Prior + scene-CV-weighted judges	0.7076	0.7510
Prior + globally-weighted judges	**0.7454**	**0.7806**

Table 1: CRAI-Bench composite Spearman (final rank: 4th). Final-system test MAE 0.1452; per-dimension test ρ: CC 0.7576, CEA 0.7502, CI 0.7084, CS 0.6916, HP 0.0000 (near-constant gold). Judge-only rows use judge outputs regenerated after a data loss; the scene-CV row keeps its original-run dev value (0.7076; it reconstructs at 0.6921 with the regenerated outputs, preserving its rank below the global weighting).

3.6 How good is the reference judge?

Because the release ships GPT-4 judgments (`gold_llm.tsv`) alongside human gold, judge reliability is directly measurable. On dev the composite correlation is $\rho = 0.249$: CEA 0.128, CC 0.121, CS 0.538, CI 0.169, HP -0.120. A judge with no access to the images (our prior, at 0.6878) outperforms it. We report this not as criticism of the benchmark (releasing the judge's raw outputs is exactly what made the measurement possible, and more benchmarks should do it) but as evidence that LLM-as-judge scores need a floor comparison before they are trusted as evaluation infrastructure (Zheng et al., 2023).

4 Task 1b: Hallucination Detection

Each item, drawn from the AynVQA benchmark's MSA track, has three statements of which exactly one is grounded. The obvious formulation, asking the model to label each statement true or false independently, permits invalid outputs (two "true"s); the official baseline reaches 0.508 combined accuracy, where combined accuracy counts an item as correct only if all three of its labels are correct. We instead ask for the *index* of the single grounded statement, in Arabic, with a JSON schema constraining the answer to $\{0, 1, 2\}$ (Appendix A). The task's semantics are fully preserved: the submission is still a complete per-statement true/false

labeling (the selected statement labeled true, the other two false), so the system does exactly what the track asks: accept the grounded statement and reject both alternatives; what changes is that the one-true structure is enforced at decode time rather than hoped for from the model. Valid labelings then hold by construction, and the official CFHR metric (of the items whose grounded statement was found, the share that still affirmed a hallucination) is exactly 0.0000 for our systems.

On the 500-item labeled dev set, `gemini-3.6-flash` and `gemini-3.1-pro-preview` tie at 0.9680 combined accuracy; on the 1,000-item blind test our submitted `gemini-3.1-pro-preview` system scores **0.964, ranking 1st**.

Isolating the constraint. To measure what the reformulation itself is worth, we ran the original formulation under matched conditions: the same model (`gemini-3.1-pro-preview`), the same images, temperature and server-side JSON schema mechanism, and an Arabic prompt identical except that it asks for an independent true/false label per statement and does not state that exactly one is true (Appendix A). On dev, this scores 0.9420 combined accuracy with a 4.0% invalid-labeling rate (of 500 items, 9 receive zero "true" labels and 11 receive two), against 0.9680 and zero for the selection decode under the same model. Two readings follow. Honestly first: measured against the official baseline of 0.508, most of the headline gap is the underlying judge, not our reformulation, since the frontier model already reaches 0.9420 under the original formulation. But the constraint still adds +0.0260 combined accuracy under a matched model, converts the 4.0% invalid tail into well-formed predictions, and guarantees CFHR $= 0$, and that +0.0260 exceeds every model-side intervention we measured: upgrading from a fast to a frontier judge is worth nothing measurable on dev, and a majority vote over three models (adding `gemini-3.5-flash` as tiebreaker) resolved the 37 test items where pro and flash disagree yet scored *lower* (0.961) despite a one-item dev advantage, an honest negative result.

Error analysis. The selection system's 16 dev errors (3.2%) are not uniform. Religion & Spirituality accounts for 6 of them, a 12.2% within-category error rate, nearly four times the overall rate; the confusions there involve fine-grained ritual and attire distinctions between visually similar practices. Errors also skew positionally: in 8 of the 16 the model wrongly selects statement 0, suggesting a mild first-position prior when evidence is ambiguous. No other category exceeds 4 errors.

One metric note: for a decoder that always emits exactly one "true", every error is a clean swap (one false statement affirmed, the grounded one rejected), so each of the official metric's instability events coincides with an accuracy error and contrastive instability (Mousi et al., 2026a) reduces exactly to $1-$accuracy (our leaderboard value: 0.036). An empty or id-mismatched submission scores a spurious perfect 0.000 on it.

5 Conclusion

Across two evaluation tracks, the pattern repeats: measure against the benchmark's own structural floor, tune judge weights against the metric that actually ranks systems rather than a proxy that cross-validation happens to measure, and let the metric's structure (ranking for Spearman, a one-of-three constraint for 1b) determine the decoding formulation. Our systems reached 0.7806 test Spearman on CRAI-Bench (4th) and 0.964 combined accuracy on Task 1b MSA (1st). The recurring lesson is that the evaluation machinery (the CV design, the metric's tie structure, the decode formulation) moved our scores more than any model choice; our concurrent ArGuard 2026 systems reached the same conclusion for threshold calibration in Arabic harmful-content detection (Younis, 2026).

Limitations

CRAI-Bench's test split is 40 instances over 8 scenes; Spearman on $n = 40$ is high-variance, our own dev bootstrap on the re-weighting gain spans zero, and small differences between systems should not be over-read. The version prior's strength is specific to benchmarks with designed caption ablations and will not transfer to naturally collected evaluation sets. Our judge ensemble is two models from one provider family, and access constraints prevented adding a third family within the evaluation window; the tuned judge weights may therefore reflect that correlated pair rather than vision judges in general. A data loss during the campaign required regenerating the judge outputs; one dev number (the scene-CV blend) is reported from the original run and does not exactly reconstruct, as disclosed in Table 1. On Task 1b we participated only in the MSA track; dialectal tracks are untested, and the 0.003 test gap between

the single judge and the three-model vote is within noise on 1,000 items.

Ethical Considerations

The CRAI-Bench and AynVQA data are released under CC BY-NC-SA 4.0 and were used for research evaluation only, with no redistribution. Cultural-accuracy scoring encodes a particular view of what counts as authentically Qatari or Gulf; our systems inherit the annotators' judgments wholesale and should not be treated as arbiters of cultural correctness. We note specifically that a system which scores cultural authenticity could be misused to police cultural expression, and that our strongest component, a frequency prior over caption versions, has no cultural understanding whatsoever, which is itself a caution against reading such scores as cultural competence.

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A Prompts

CRAI judge rubric (verbatim; the judge receives the reference image, the generated image, and this text with the caption inserted):

```
You are evaluating whether an AI-GENERATED image faithfully represents Qatari/Gulf Arab culture, using a REFERENCE photograph of the authentic scene and the generation caption. The FIRST image is the authentic REFERENCE. The SECOND image is AI-GENERATED from this caption: "{caption}". Score the GENERATED image on four dimensions in [0,1] (higher = better), then list hallucinated elements: 1. CEA (Cultural Element Accuracy): are the culturally specific elements (garments like thobe/abaya/batoola, architecture, objects) depicted correctly compared to the reference? 2. CC (Cultural Completeness): how many of the culturally required elements implied by the caption and visible in the reference are present at all? 3. CS (Cultural Style): does the overall style/setting read as authentically Qatari/Gulf? Answer with exactly one of: 0, 0.25, 0.5, 0.75, 1. 4. CI (Caption-Image alignment): does the generated image match what the caption asked for? 5. HALLUCINATIONS: list each concrete element in the generated image that is culturally WRONG or fabricated (wrong garment type, invented patterns, non-Gulf architecture, anachronisms). Empty list if none. Only name things you can point to in the image. Return ONLY JSON: {"CEA": <float>, "CC": <float>, "CS": <one of 0|0.25|0.5|0.75|1>, "CI": <float>, "hallucinated_elements": ["...", ...]}
```

The rubric's short dimension glosses are ours; scoring uses the official composite weights

of §3.1 with HP computed as $\min(0.1 \times |\text{hallucinated_elements}|, 1)$.

Task 1b selection prompt (submitted system). The prompt is in Arabic; we give an English gloss here and release the verbatim Arabic with the code: "Look at the image carefully. You have three statements about the image; only one of them is correct and actually grounded in what the image shows, and the other two are wrong despite appearing culturally plausible. [Statements 0, 1, 2.] Identify the single correct statement. Answer in JSON only: {reasoning, grounded_index $\in \{0, 1, 2\}$}."

Task 1b true/false ablation prompt. Identical in language and structure, except that it does not state that exactly one statement is true and asks: "For each statement separately, decide whether it is true and actually grounded in what the image shows, or false. Answer in JSON only: {reasoning, label_0, label_1, label_2}" with three boolean fields in the schema.

B Blend procedure

For each dimension d: (1) compute $r = (1 - w_d) \cdot \text{rank}(\text{prior}_d) + w_d \cdot \text{rank}(\text{mean of the two judges}_d)$ over the split being predicted; (2) min-max normalize r to $[0, 1]$ and map it linearly into the $[p_5, p_{95}]$ range of dimension d's train gold, clipping to $[0, 1]$; (3) snap CS to the nearest legal tier; (4) recompute the composite with the official weights. The weights $w_d \in \{0, 0.1, \ldots, 1\}$ are selected by two rounds of coordinate descent maximizing global composite Spearman on train gold only. For the test submission the prior is refit on train+dev. Priors use (version, category) means with a version-only fallback for unseen categories.